VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Data Storage MPU

AFLCMC/WA

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1.0 Overview

Data Storage is a Level 2 Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). Data Storage MPUs may provide encrypted and secure data storage accessible to specific AMS GRA elements at the Mission System Integrators (MSIs) discretion through the Abstract Service Bus (ASB) and/or Avionics Backplane Bus (ABB). Data Storage MPUs can be housed within a Processor Chassis and Backplane (PCaB) or be standalone units.

1.1 Purpose

The Data Storage MPU Volume introduces the Data Storage MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features provide design guidance on what is typically expected for the Data Storage MPU. The interface diagram contains a graphical representation of the types of interfaces that can exist (from the perspective of the Data Storage MPU). The interfaces table contains a textual representation of the interfaces and their descriptions. Compliance requirements describe what is required for the Data Storage MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The Data Storage MPU Volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, MSI, or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the Data Storage MPU Volume.

Standard	Title	Version/Date
AMS GRA Models	AMS GRA Architecture Model	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Data Storage MPU.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Data Storage MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	PCaB MPU Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Data Storage MPU and Related Entities.

2.0 Data Storage

The Data Storage MPU provides the functionality needed by a Mission System to store and protect sensitive AMS GRA software and Data at Rest (DaR). The Data Storage MPU may implement cryptographic mechanisms to support the bulk encryption of stored DaR as data. This bulk encryption functionality may enable the MPU to be at rest.

The data securely stored within the Data Storage MPU is accessible by specific AMS GRA architectural elements after obtaining any keys required for decryption. The Data Storage MPU may have the capability to support the secure start-up, secure deployment of software, data sanitization and zeroization, and the secure shutdown of the system.

2.1 Key Features

Table 2.1.1 identifies the key features of Data Storage MPUs and the benefit of these features. This table is not prescriptive but identifies a number of attributes that AMS GRA definitions and interfaces are designed to support. This table is included in the Data Storage MPUs documentation template so that Data Storage MPUs providers can summarize the specific features supported by a Data Storage MPUs instantiation.

Key Feature	Benefit
Data Encryption/Decryption	Provides DaR protection through confidentiality and integrity protection measures.
Data Storage Management Tool	Enables widespread interoperability and market competition using an open commercial standard interface called the Network File System (NFS).

Table 2.1.1: The Data Storage Key Features Table Describes the Specific Functionality Supported by a Data Storage Instantiation.

2.2 Interface Diagram

Figure 2.2.1 and **Figure 2.2.2** depict the Data Storage MPU and its external interfaces for the PCaB configuration and standalone configuration, respectively. Unlike the standalone configuration, Test/Debug in the PCaB configuration is the only interface that can provide a direct connection from Data Storage MPUs to the platform. The PCaB and standalone interface diagrams do not represent mutually exclusive MPU configurations. The interface diagrams only contain logical entities. Interface connections denoted as pass-through only represent situations where the sending MPU is not the original source and does not use or modify what is provided to the receiving MPU.

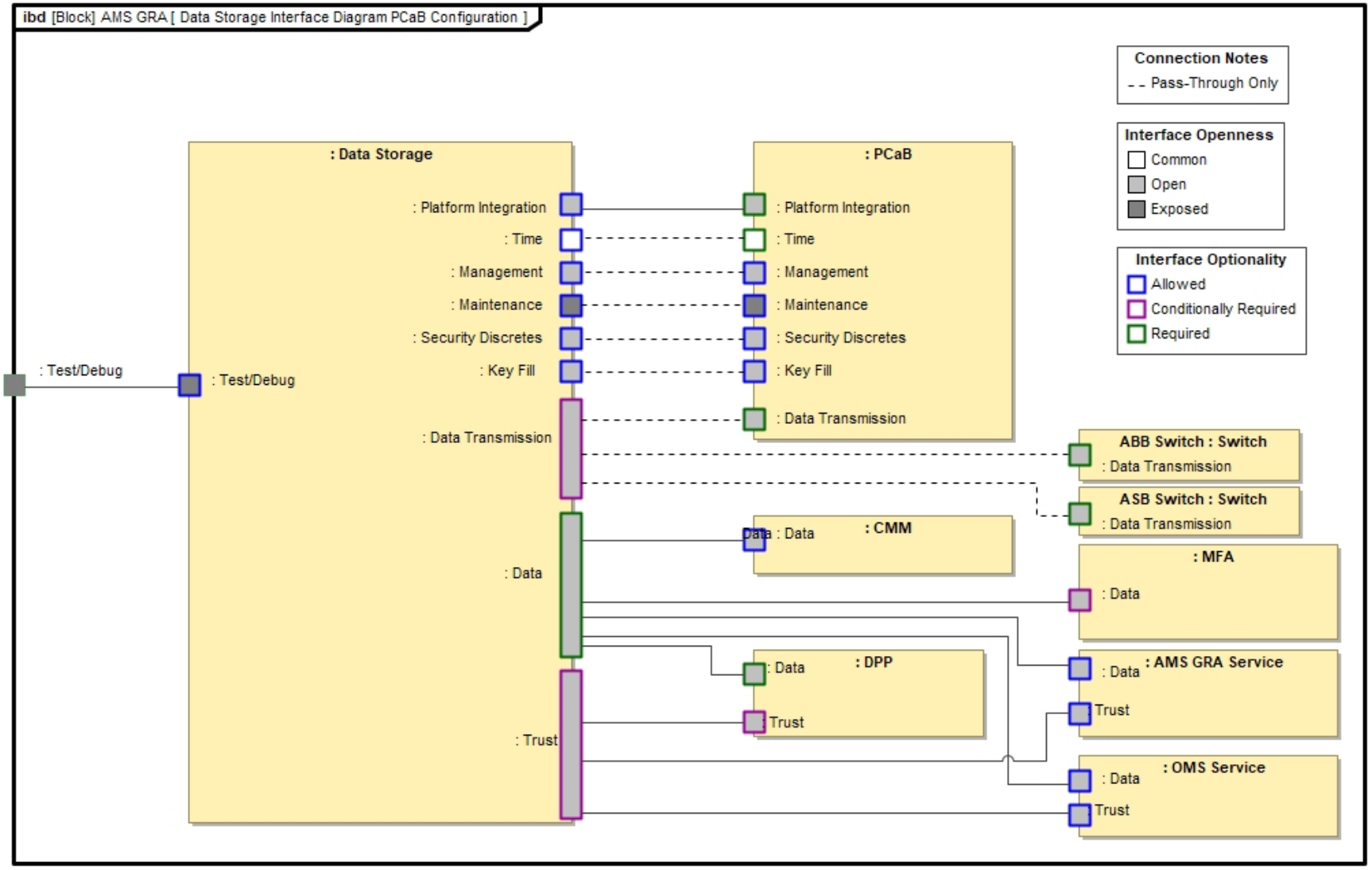


Figure 2.2.1: This Data Storage MPU Interface Diagram Shows Interfaces on the Data Storage MPU and Their Connections When the MPU Is Hosted on a PCaB.

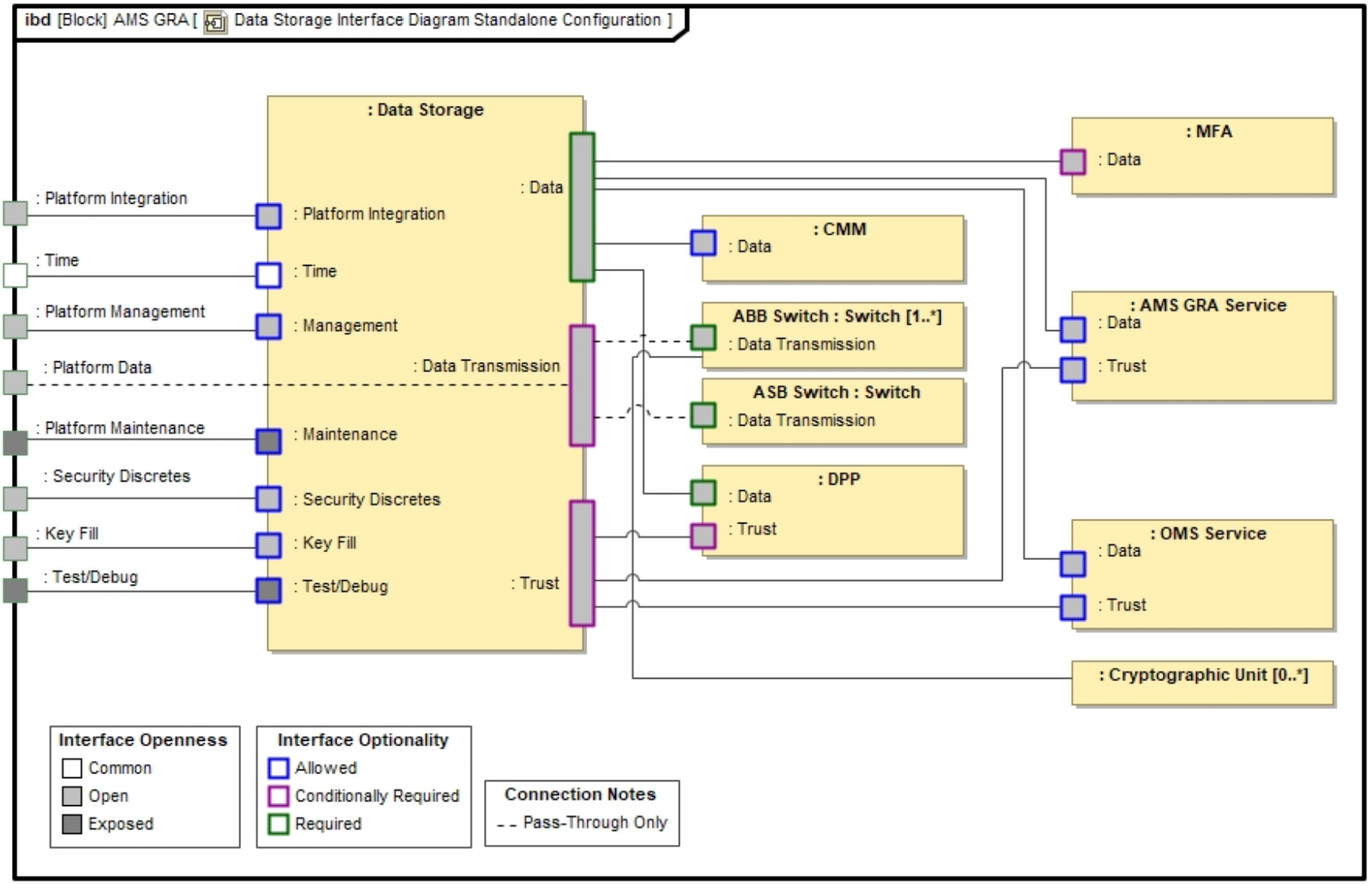


Figure 2.2.2: This Data Storage MPU Interface Diagram Shows Interfaces on a Standalone Data Storage MPU and their Connections. This Configuration Allows for Other Interfaces to Have Direct Connections to the Platform in Addition to Test/Debug.

2.3 Interface Table

Table 2.3.1 identifies the Data Storage MPUs interfaces depicted in **Figure 2.2.1** and **Figure 2.2.2**. When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally Required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

Interface	Documentation	Interface Optionality	Condition
Trust	The Trust interface enables secure connectivity between endpoints using cryptographic means to establish a secure channel. This channel may be used to verify integrity, authenticate endpoints, or to transfer data requiring higher levels of assurance than normal data transfers, such as CPI, or Software. The implementation of this interface may vary depending on assurance level, exportability, and the various mission specific requirements of the adopting program.	Conditionally Required	Required if the MPU is used to transfer mission data, software, or associated Key Material, and there are no other interfaces implemented which support integrity verification.
Data	The Data interface represents the ability to exchange digitized information between the Data Storage MPU and other MPUs.	Required	N/A
Data Transmission	Data Transmission represents the interface support used to facilitate a data transmission connection, independent of the type of data that is being transmitted. The connections from Data Transmission to the other MPUs denote which MPUs can be connected for transmitting data.	Conditionally Required	Required when Data Storage connects to Platform Data and transfers data.
Key Fill	The Key Fill interface is used to load keys into the system to support encryption across the architecture. This interface provides a dedicated path to load a master key or keys that then support key derivations to be used by other MPUs to encrypt communication or for other cryptographic functions.	Allowed	N/A

Interface	Documentation	Interface Optionality	Condition
Maintenance	The Maintenance interface provides the capability to perform preventative and corrective maintenance while installed on the platform.	Allowed	N/A
Management	The Management interface provides the capability to support management of the MPU in an AMS GRA compliant mission system. Example management for the Data Storage includes centralized control and monitoring (e.g. IPMI).	Allowed	N/A
Platform Integration	The Platform Integration interface represents the general physical interfaces that an MPU will typically have with the host platform, including but not limited to mounting, electrical, hydraulic, and cooling connections.	Allowed	N/A
Security Discretes	The Security Discretes interface specifies the security interfaces between the Data Storage and the platform.	Allowed	N/A
Test/Debug	The Test/Debug interface specifies an interface to use for test equipment purposes.	Allowed	N/A
Time	The Time interface specifies the interface to an AMS GRA prescribed time source.	Allowed	N/A

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 states the applicable MPU compliance requirements for the Data Storage MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

See the Mission System Volume for additional details about MPU Fault Reporting.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-005	An MPU with CPI on a Trusted Ops platform shall cryptographically protect CPI using an open implementation.	Tier 3	Analysis	
GRA-MPU-006	An MPU that processes or decrypts CPI on a built-for-export platform shall implement the OMS Standard.	Tier 2	Analysis	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-037	An MPU shall use, at a minimum, NFS as a protocol for the Data interface.	Tier 1	Inspection	
GRA-MPU-038	An MPU shall use open, commercial standards for the Management interface.	Tier 1	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-047	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the execution environment.	Tier 2	Inspection	
GRA-MPU-048	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the build environment.	Tier 2	Inspection	
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-061	An MPU shall ensure the authenticity and integrity of MPU software and firmware during MPU initialization.	Tier 1	Inspection	
GRA-MPU-062	An MPU shall support integrity verification to the Mission System over a secure communication channel.	Tier 1	Demonstration	
GRA-MPU-063	An MPU shall implement secure communications using open standards.	Tier 1	Analysis	
GRA-MPU-064	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of a software load when received over a non-secure communication channel.	Tier 1	Demonstration	
GRA-MPU-065	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of software locally stored at rest, not including bootloaders; prior to updating, loading, and executing.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-067	An MPU shall publish details of security relevant events in an open format.	Tier 1	Demonstration	
GRA-MPU-074	An MPU that processes or decrypts CPI on a built-for-export platform shall fully implement OMS Standard.	Tier 3	Analysis	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-084	An MPU that implements the Time interface shall use one or more of PTP, and OMS legacy timing.	Tier 1	Inspection	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUs and CDS MPUs report faults as defined in the CDS and CU IDD. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-088	An MPU shall report health and status on its Management Interface.	Tier 1	Demonstration	
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	Tier 1	Inspection	Note: MFA reports BIT information via the MEL interface

Table 3.0.1: A Data Storage MPU's Adherence to the Compliance Requirements Determines its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
ASB	Abstract Service Bus
BIT	Built-In Test
CDS	Cross Domain Solution
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DK	Development Kit
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
IPMI	Intelligent Platform Management Interface
MBSE	Model-Based Systems Engineering
MEL	Multi-Function Aperture (MFA) Encapsulation Layer
MFA	Multi-Function Aperture
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NFS	Network File System
NSA	National Security Agency
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PTP	Precision Time Protocol

RoT	Root of Trust
SAST	Static Application Security Testing
SK	Starter Kit
UCI	Universal Command and Control Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revisions adds prose updates and clarifications.	July 24, 2025	SK13.0
B	This revision adds several new requirements for the Data Storage MPU and references to the Cryptographic Unit (CU). Adds new requirements GRA-MPU-088 and GRA-MPU-91.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Data Storage MPU Volume Between Documentation Releases.